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IMPLIED VOLATILITIES AND ARBITRAGE OPPORTUNITIES IN THE ITALIAN OPTIONS MARKET

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SUMMARY

This paper, which follows up the analyses of an earlier study published in December 1988, starts by updating the description of the regulatory framework for Italian stock options (premium contracts), with reference both to the regulations issued by the Consob (the Italian SEC) and the changes in tax law. This is followed by figures on the size of the market and its development in recent years. After describing the standards operators use to value premium contracts and quantifying the deviations from the theoretical values, we examine the techniques for forecasting share price volatilities and verify the information content of the implied volatilities of premium contracts. The last part of the paper describes an extensive simulation designed to reveal arbitrage opportunities that operators failed to exploit during the period in question, with account being taken of transaction costs.

CONTENTS

1. INTRODUCTION

The markets for share premium contracts on the leading Italian stock exchanges[1] correspond to the stock option markets found in Anglo-Saxon countries, though they differ from these in several respects. In the first place, premium contracts are options on forward contracts and not on spot contracts. Furthermore, the premium is not paid when the contract is written, but on the settlement day of the forward contract on which the option is written. As on Anglo-Saxon markets, the option that can be exercised under a premium contract is of the American type since it can be exercised before the expiration date *(risposta premi* day).[2]

The present study, which is a follow-up of an earlier paper, starts with a description of the current legal framework of Consob regulations and tax law (§2). The statistics on the size of the market are then updated and the sample used for the analysis described (§3). After briefly discussing the formula for valuing *dont* contracts (equivalent to call options) and the features of the replicating portfolio (§4), the study focuses on the optimum forecast of share price volatility (§5). The last section presents a simulation designed to determine whether arbitrage opportunities exist that are not exploited by operators (§6).

2. THE ITALIAN OPTIONS MARKET: REGULATORY ASPECTS

2.1 Striking prices

Consob Resolution no. 4764 of July 10, 1990 made new changes in the rules on the striking prices of premium contracts (Table 1). There are still seven price ranges, of which three are unchanged and three wider than before. The changes have made the grid of striking prices tighter.

2.2 Margin requirements

In the same Resolution the Consob streamlined the regulations governing the margins required in respect of single and double premium contracts,[3] thereby making arbitrage operations easier (Table 2). This change was the result of the rescission of the Resolution adopted by the Consob on April 2, 1986, which forbade short sales of premium contracts.

The new regulations came into effect at the start of the September 1990 stock exchange month (90/8/17).[4]

2.3 The taxation of stock exchange contracts

Decree Law no. 70 of March 14, 1988, ratified with amendments as Law no. 154 of May 13, 1988, considerably simplified the taxation of stock exchange contracts. The rates applied now depend only on the legal forms of the parties and the type of security involved, and no longer vary with the type of contract [cash, forward or contango (*riporti*)] or its maturity (Table 3).

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1. The Milan and Rome Stock Exchanges have long had special pits for the trading of premium contracts and Consob Resolution no. 4435 of 90/1/9 authorized the creation of another on the Turin Stock Exchange.

2. Options have to be exercised by 10.00 on the day fixed by the stock exchange calendar. In the event of an option not being exercised, the contract has to be executed to the best advantage of the buyer, with reference to the list price of the day before *risposta premi* day.

3 Double premium contracts (*stellage*, *strip* and *strap* contracts) represent various combinations of single premium contracts (*dont* and *put*).

4 The 10 July Resolution was subsequently amended by urgent order no. 35 of August 22, 1990 (approved in Resolution no. 4849 of August 27, 1990), which, among other things, made it compulsory, from the date the order was issued, for the execution of simple and double premium contracts involving the delivery of securities to be preceded by the delivery of the securities (by the third trading day after the execution of the order). This measure remained in force until October 2, 1990.

TABLE 1 Striking prices (lire)

Price range				Reference value	Striking price variation
	up	to	500	0	10
from	501	to	1,000	500	25
from	1,001	to	5,000	1,000	50
from	5,001	to	15,000	5,000	100
from	15,001	to	50,000	15,000	250
from	50,001	to	100,000	50,000	500
		more than	100,000	100,000	1,000

TABLE 2 Margin requirements for premium contracts (Consob resolution no. 4764 of July 10, 1990)

MARGIN FOR SINGLE CONTRACTS
(*DONT/PUT*)

DONT

Buyer: has to deposit the premium. Subsequent forward sales for settlement on the same day are subject, up to an equal quantity, to the requirement that the buyer shall deposit the amount of any positive difference between the striking price and the forward sale price.

Seller: EITHER

has to demonstrate the availability of the securities. If he does not expect to have to deliver the securities, he can sell them under a forward contract or a premium contract, and deposit the striking price.

OR

has to deposit the striking price.

PUT

Buyer: has to deposit the premium. Subsequent forward purchases for settlement on the same day are subject, up to an equal quantity, to the requirement that the buyer shall deposit the amount of any positive difference between the forward purchase price and the striking price.

Seller: has to deposit the striking price.

MARGIN FOR DOUBLE CONTRACTS
(*STELLAGE/STRIP/STRAP*)

Buyer: has to deposit the premium. Subsequent forward purchases/sales for settlement on the same day are not subject, up to an equal quantity, to the margin requirements.

Seller: EITHER

has to demonstrate the availability of the securities sold under the premium contract and deposit the striking price of those purchased. If he does not expect to have to deliver the securities, he can sell them under a forward contract or a premium contract, and deposit the maximum commitment under the contract.

OR

has to deposit the maximum commitment under the contract.

N.B. single premium contracts linked to earlier single premium contracts with the same striking price (or a higher striking price if *dont* contracts, or a lower striking price if *put* contracts), or to forward contracts or to contango (*riporti*) contracts of the opposite sign are not subject to a margin requirement up to the same quantity if they are for settlement on the same day. The same applies to double premium contracts linked to earlier double contracts of the opposite sign for settlement on the same day. Nor is margin required for the "closed transformations" (combinations of forward and premium contracts that fully hedge the commitments entered into) operators undertake for the same settlement day. In the event of closed transformations extending into other accounts, only the forward sales and purchases involved are subject to a margin requirement.

TABLE 3 Taxation of stock exchange contracts (rate per 100,000 lire or fraction thereof)

Parties	Securities		
	Shares	Foreign currencies[1]	Bonds[2]
Individuals	140	100	16
Banks and other[3]	50	90	9
Stockbrokers	8	40	9

1. Excluding cash contracts.
2. Including repurchase agreements. The tax liability cannot exceed 1,800,000 lire, except for contracts between individuals.
3. Including contracts concluded with the intervention of stockbrokers, registered banks and commission agents.

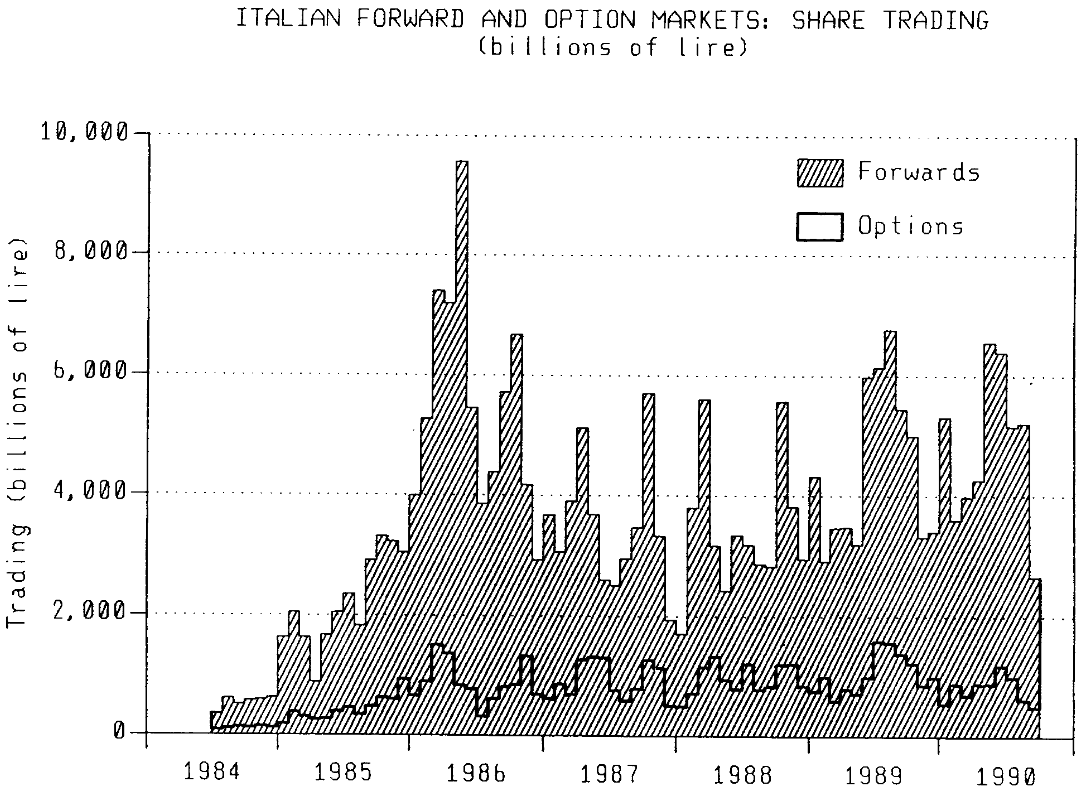

Figure 1

Accordingly, forward, premium and contango (*riporti*) contracts that are entered into on the stock exchange through a broker are taxed at the rate of .005 per cent (previously .01 per cent).[5]

3. THE SIZE OF THE MARKET AND THE RETURN ON DONT CONTRACTS

3.1 The size of the market and the sample used

The size of the market

Between 1985 and 1990 the volume of premium contracts written has varied considerably, with alternating periods of expansion and decline (Table A1).[6]

As a percentage of forward sales, the value of the shares underlying premium contracts ranged from a minimum of 8 per cent in July 1986 to a maximum of 50 per cent in June 1987, and amounted to 17 per cent in the first nine months of 1990 (Figure 1).

The value of premium contracts has nonetheless remained small, amounting to 263 billion lire in 1985, to 340 billion in 1989 and to 174 billion in the first nine months of 1990.

The sample

The sample used in the following analyses comprises the *dont* contracts written on the Milan Stock Exchange in the 24 stock exchange months from 17 August 1988 to 16 August 1990 (505

[5] The basic provisions on the taxation of stock exchange contracts (Royal Decree 3278/1923, as amended) require that the stamped contract note shall indicate the names of the parties, the date by which it is to be executed, the date and the details of the transaction. The tax liability on stock exchange contracts can also be settled by making quarterly installment payments to the registry office without using stamped contract notes.

[6] The Tables numbered A... are in Appendix A.

trading days). The information published in the Listino Ufficiale of the Milan Stock Exchange on the *dont* contracts written each day is classified by striking price and maturity and includes the minimum and maximum premia, the number of contracts written and the number of underlying shares.[7]

In addition to this information, account is taken of the average offset fixed by the Stockbrokers' Executive Committee in the case of rights issues, dividends detached, share splits and mergers in order to make the necessary changes to the striking prices of outstanding premium contracts. If a subscription (or bonus share) right is detached during the life of the contract, the striking price is reduced by the average offset, and in the event of a dividend payment, by the gross dividend.[8] If a stock undergoes a split (or reverse split) or swapped for another following a merger, the striking price is divided by the ratio of the split (or reverse split) or by that of the swap, in other words by the ratio of the number of new shares to the number of old shares given in exchange; the number of underlying shares is modified by multiplying the original number by the same ratio.

For each of the 505 trading days in the sample, the official closing prices of the 205 shares on which *dont* contracts were written are also observed, giving more than 100,000 observations.[9] Finally, the contango (*riporti*) rates are observed for the underlying securities of premium contracts written for the end of the subsequent stock exchange month (*fine prossimo*).[10]

Analysis of the data

During the period in question, 19,430 *dont* contracts were written, of which 71.6 per cent were *per fine prossimo*.[11] The average maturity at issue amounted to 34 days. When they were written, the *dont* contracts were generally slightly out of the money: the average difference between the striking price and the forward price for the end of the current month amounted to 1.0 per cent of the latter. In 9.1 per cent of the cases considered (1,774 contracts), the original terms of the contract were modified as a result of operations involving the capital of the company or the distribution of dividends. More than 23 per cent of the premium contracts in the period were written on just one stock (FIAT ordinary; Figure 2).

On average 38 different contracts were written each trading day, with a minimum of 13 on October 16, 1989 and a maximum of 73 on June 14, 1989. The number of striking prices per day for the same share averaged 1.6, with a maximum of 7 for contracts written on FIAT ordinary shares on June 9, 1989 (3 for the end of the current month and 4 for the end of the subsequent month).

3.2 The distribution of returns

The range of the rates of return[12] that can be obtained with the purchase of a *dont* contract is boundless, both to the left (-∞) and to the right (+∞), as in the case of forward purchases of shares. However, in contrast with forward purchases, the distribution of the returns obtainable by purchasing

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[7] The same information is available for *put, stellage, strip* and *strap* contracts. This is a considerable improvement on the past [see Barone and Cuoco (1988, footnote 6, p. 45)]. However, the statistics still do not include the weighted averaged premium, the most important indicator for operators in a continuous market such as that for premium contracts.

[8] Article 41 of the Milan Stock Exchange Rulebook lays down that "For premium contracts involving underlying securities for which subscription rights are exercised before *risposta premi* day, the price is reduced by the average offset fixed by the Stockbrokers' Executive Committee for the subscription rights. The premium contract option is exercised and settlement made for *ex rights* shares. For dividends, bonus shares, *riporto* rights and reimbursements, the premium contract option is exercised and settlement made with the striking price reduced by the gross amount. The premium remains unchanged." See also Barone and Cuoco [1988; pp. 13-15 and pp. 21-22].

[9] In what follows it is assumed that the terms of the premium contracts are observed at the same time as the official closing prices of the corresponding forward contracts were fixed.

[10] Premium contracts for the current stock exchange month are less common, while virtually no use is now made of those with longer maturities (*fine secondo mese*, etc.).

[11] All the *dont* contracts written on a security during the same trading day, and with the same striking price and maturity, were considered as a single contract.

[12] The rates of return are calculated as $R_T = \ln[\mathrm{Max}(F_T - K,0)/P_D]$ and not as $R_T = [\mathrm{Max}(F_T - K,0) - P_D]/P_D$, which are bounded on the left (–100%) (in fact, the buyer's liability is limited by his not being able to lose more than the premium). It is also assumed that the option is not early exercised and that transaction costs are zero.

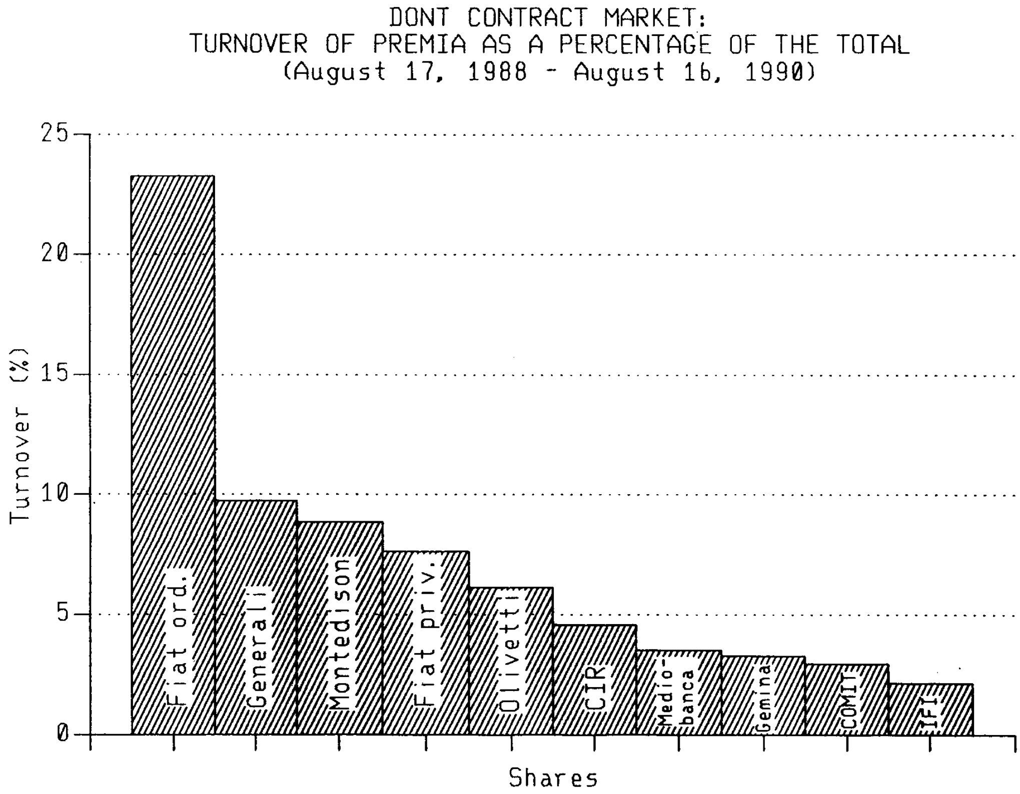

Figure 2

premium contracts is highly asymmetrical: the probability of losing all the capital invested is high, since it is sufficient for the price of the underlying securities to be less than the striking price on *risposta premi* day, while the probability of arriving at the opposite tail of the distribution (infinite gain) is negligible. On the other hand, since the capital invested consists only of the premium, the probability of obtaining very high positive yields is greater than for forward purchases.

The distribution observed in the two-year period considered is shown in Figure 3.[13] For 76 per cent of the contracts observed, buyers incurred a loss, with the total capital invested being wiped out for two contracts out of three. On the other hand, 8 per cent of the contracts generated (continuously compounded) returns of more than 100 per cent, with a maximum of 343 per cent.[14]

4. THE THEORETICAL VALUE OF PREMIA

4.1 A valuation formula

Barone and Cuoco (1988) have shown that under hypotheses similar to those of Black-Scholes,[15] the

[13] The distribution has been weighted by attributing to each contract a weight corresponding to the value of the premium involved.

[14] The highest return was earned by buying a *dont* contract on Pacchetti shares on March 13, 1989 for 5 lire with a striking price of 480 and maturity on April 11, 1989. Since the price of the underlying share increased from 466 to 635 lire, the return of the investment was equal to 30 [ln(30)=3.43].

[15] According to the formula proposed by Black-Scholes (1973), the value of a European call option, with striking price K and maturity T, is given by $c = S_tN(d_1) - e^{-rT}KN(d_2)$, where $d_1 = [\ln(S_t/K) + (r + \frac{1}{2}\sigma^2)T]/(\sigma\sqrt{T})$ and $d_2 = d_1 - \sigma\sqrt{T}$. S_t is the spot price of the underlying security, r is the (continuously compounded) risk-less interest rate, and σ is the standard deviation of the instantaneous rate of change of S_t.

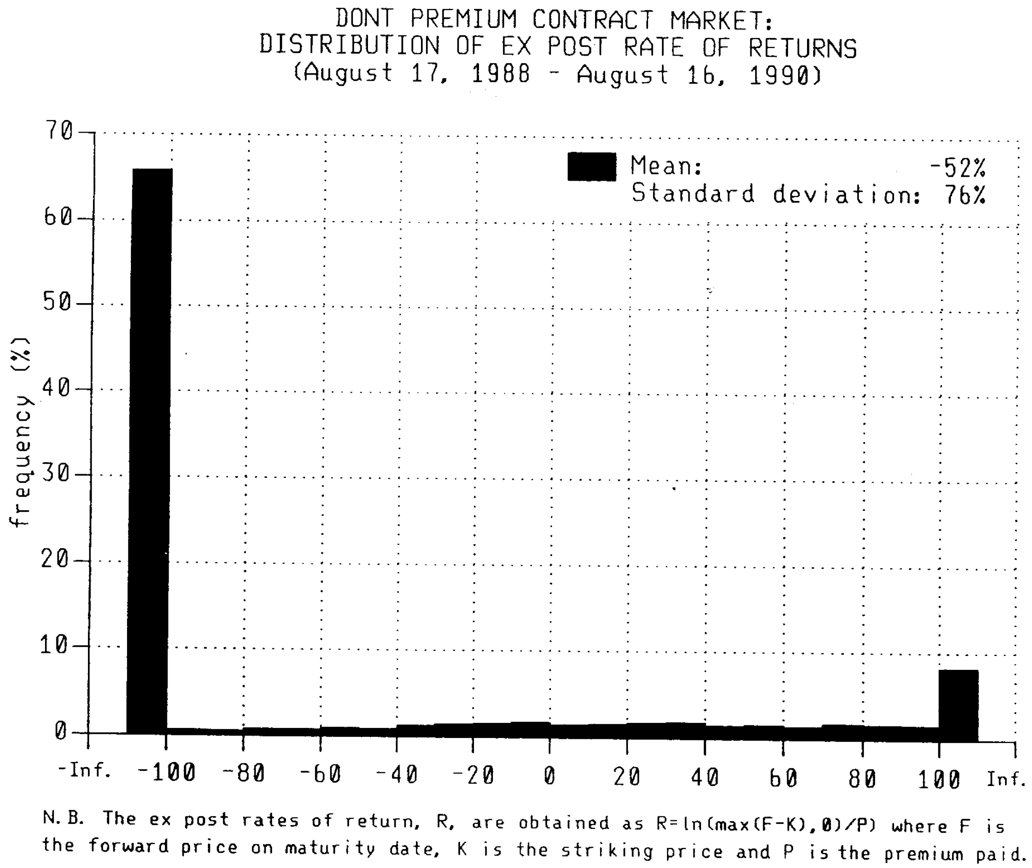

Figure 3

theoretical value of a *dont* contract[16] is given by:

$$\hat{P}_D(t,T,\tau) = F(t,\tau)N(d_1) - KN(d_2) \tag{1}$$

where:

- $\hat{P}_D(t, T, \tau)$ is the value, at time t, of the *dont* contract maturing at time T for settlement at time τ;
- $d_1 = [\ln(F/K) + \frac{1}{2}\sigma^2 T]/(\sigma\sqrt{T})$ and $d_2 = d_1 - \sigma\sqrt{T}$;
- $N(\cdot)$ is the standardized normal distribution function;[17]
- $F(t, \tau)$ is the forward price of the underlying share, observed at time t, for settlement at time τ;[18]
- K is the striking price of the premium contract;
- T is the maturity of the premium contract (*risposta premi* day);
- τ is the settlement date;
- σ is the volatility of the price of the underlying share.

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[16] The value of the *put* contract and of compound contracts (*stellage*, *strip* and *strap*) can easily be expressed in terms of the value of the *dont* contract [Barone and Cuoco (1988; p. 23)].

[17] $N(\cdot)$ can be calculated using the following polynomial approximation [Abramowitz (1970)]:

$$N(z) \approx 1 - (1/\sqrt{2\pi})e^{-z^2/2}\,(b_1k + b_2k^2 + b_3k^3 + b_4k^4 + b_5k^5)$$

where: $k \equiv 1/(1 + .2316419z)$ $b_1 = .319381530$ $b_2 = -.356563782$ $b_3 = 1.781477937$ $b_4 = -1.821255978$ $b_5 = 1.330274429$.

[18] $F(t, \tau)$ is thus the ordinary forward price (which is not always observable) with the same settlement date as the premium contract.

TABLE 4 The dont contract with the extreme values of the explanatory variables

Extreme values	Variables			
	F	K	T	σ
0	0	F	$\begin{cases} 0 & \text{if } F < K \\ F-K & \text{if } F > K \end{cases}$	$\begin{cases} 0 & \text{if } F < K \\ F-K & \text{if } F > K \end{cases}$
∞	∞	0	F	F

The *dont* contract is thus a function of F, K, T and σ;[19] Table 4 shows how it varies with the extreme values of these four variables.

In particular, it is worth noting that when the striking price is zero ($K = 0$), the theoretical value of the *dont* contract given by formula (1) is equal to the forward price, F. In this case, in fact, the *dont* contract is equivalent to a forward contract, since it will always be advantageous to exercise the right to buy the underlying securities.[20]

As regards the sensitivity of the *dont* contract with respect to the four variables, the following expressions apply:[21]

$$\Delta = \frac{\partial \hat{P}_D}{\partial F} = N(d_1) > 0 \tag{2}$$

$$-\Theta = \frac{\partial \hat{P}_D}{\partial T} = \frac{FN(d_1)\sigma}{2\sqrt{T}} > 0 \tag{3}$$

$$\frac{\partial \hat{P}_D}{\partial K} = -N(-d_2) < 0 \tag{4}$$

$$\frac{\partial \hat{P}}{\partial \sigma} = FN'(d_1)\sqrt{T} > 0 \tag{5}$$

where $N'(\cdot)$ is the standardized normal density function. Furthermore,

[19] In contrast with the Black-Scholes formula, formula (1) does not include the interest rate, which is implied in the price of the forward contract $F(t, \tau)$. In fact, we have $F(t, \tau) = S_t\exp[r_{rip}(\tau - t)]$, where $S_t = F(t, t)$. The *riporti* rate, r_{rip}, is equal to the difference between r, the interest rate on money loans, and r_a, the rate on loans of securities.

[20] In the same way, putting $K = 0$ in the Black-Scholes formula results in the theoretical value of the call option being equal to the spot price S. In this case, however, the two contracts are not always equivalent since the spot contract provides for the immediate availability of the stock, while with a European call option delivery is delayed until the maturity of the option. The Black-Scholes formula thus presupposes that the current rate for the loan of securities is zero or, in other words, that the market is indifferent as to whether it holds securities today or tomorrow.

[21] For example, to calculate $\partial \hat{P}_D / \partial \sigma$, we have

$$\begin{aligned}\frac{\partial \hat{P}_D}{\partial \sigma} &= FN'(d_1)\frac{\partial d_1}{\partial \sigma} - KN'(d_2)\frac{\partial d_2}{\partial \sigma} \\ &= FN'(d_1)\frac{\partial d_1}{\partial \sigma} - KN'(d_1)\frac{F}{K}\left(\frac{\partial d_1}{\partial \sigma} - \sqrt{T}\right) = FN'(d_1)\sqrt{T}\end{aligned}$$

since

$$N'(d_2) = \frac{1}{\sqrt{2\pi}}e^{-½d_2^2} = N'(d_1)\frac{F}{K}.$$

$$\Gamma = \frac{\partial \Delta}{\partial F} = \frac{N'(d_1)}{F\sigma\sqrt{T}} > 0 \tag{6}$$

$$\Omega = \frac{\Delta F}{\hat{P}_D} = \frac{N(d_1)F}{\hat{P}_D} > 0 \tag{7}$$

4.2 The replicating portfolio

The value of a *dont* contract written at time t, with maturity at time T and settlement at time τ, is equivalent to a dynamically rebalanced portfolio acquired by buying $N(d_1)$ shares forward for the same settlement date[22] and making forward sales (purchases) of bonds amounting to $P_D - \hat{P}_D > 0$ ($P_D - \hat{P}_D < 0$) [Barone e Cuoco (1988, §4.1)].[23]

In the same way as when a *dont* contract is bought, the acquisition of the replicating portfolio does not involve any outlay before the settlement date. The proportions of the portfolio must be adjusted continuously as F varies, so as to ensure that its value is always equal to that of the *dont* contract. If the *dont* contract expires in-the-money [$F(T, \tau) > K$; $T = 0$], the replicating portfolio will be found to comprise a share bought forward [$N(d_1) = 1$] and a bond sold (bought) forward [$N(d_1) = N(d_2) = 1$], amounting to $P_D - [F(T, \tau) - K]$: the value of the portfolio will be equal to $F(T, \tau) - K - P_D$. If, on the other hand, the *dont* contract expires out-of-the-money ($F(T, \tau) < K$; $T = 0$), the replicating portfolio will comprise zero shares bought forward [$N(d_1) = 0$] and a bond sold forward [$N(d_2) = 0$], amounting to P_D: the value of the portfolio will thus be $-P_D$.

Of the four variables in formula (1), two (K and T) are left, within certain limits,[24] to the counterparts to determine, while the other two (F and σ) have to be measured. The forward price, $F(t, \tau)$, is normally observable for *fine corrente* premium contracts ($\tau = \tau_1$) and can be estimated accurately in the case of *fine prossimo* contracts ($\tau = \tau_2$). By contrast, it is not easy to estimate σ.

In order to determine the *fine prossimo* forward price, $F(t, \tau_2)$, knowing the *fine corrente* forward price, $F(t, \tau_1)$, it is necessary to know the (continuously compounded) *riporti* rate (r_{rip}) and to use the following expression:[25]

$$F(t, \tau_2) = F(t, \tau_1)\exp\left[r_{rip}(\tau_2 - \tau_1)\right] \tag{8}$$

In order to determine r_{rip}, which is not observable every day, some operators refer to the half-sum of the minimum and maximum *riporti* rates applied by banks and published after the last *riporti* meeting. Instead, in this study we have used the *riporti* rates observed at the date τ ($t \geq \tau$), corresponding

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[22] The proportion of shares in the replicating portfolio, $N(d_1)$, is known as the hedge ratio.

[23] The forward sale of bonds is equivalent to the opening of a line of credit to be used at the time of settlement and corresponds to the need to ensure the availability at time τ of the amount required to pay the contractual premium, net of the gain expected from the exercise of the *dont* option.

[24] T must coincide with one of the *risposta premi* dates fixed in the stock exchange calendar established every year by the Consob and K must correspond to one of the values laid down in the grid of striking prices that the Consob also set up.

[25] *Fine prossimo* purchases and sales are not symmetrical operations in terms of the margin required when the *riporti* contract that carries the position forward to the subsequent stock exchange month is written. The *fine prossimo* buyer is treated differently from the *fine prossimo* seller: the *riportato* (the *fine prossimo* buyer) loans the securities bought *per fine prossimo* and receives a money loan amounting to 50 per cent of their value (at the average end-of-account price, known as the *prezzo di compenso*), while the *riportatore* (the *fine prossimo* seller) lends money and receives securities amounting to 200 per cent of the money loan. Accordingly, at the current settlement the *riportato* will be debited the agreed amount and credited 50 per cent of the value (at the *prezzo di compenso*) of the securities lent, while the *riportatore* will be credited the agreed amount and debited 50 per cent of the value (at the *prezzo di compenso*) of the securities borrowed.

to the last *riporti* meeting;[26] for the securities not involved in *riporti* operations, the maximum observed rate has been assumed to apply.[27]

4.3 Comparison with the Black-Scholes formula

Before addressing the question of estimating r in the next section, we shall compare the results obtained, for given values of σ, using formula (1) and two methods that are used in practice. Some operators, for example, use the Black-Scholes formula without modifying it in any way, while others determine the value of the *dont* premium by multiplying the striking price by a percentage that varies with demand.[28]

Table A2 shows the theoretical values of the *dont* premium obtained for various values of F, K, T e σ, using formula (1) and the Black-Scholes formula. For the sake of comparison, we also show the ratios of the theoretical values (net of any positive difference between the current price and the striking price) and the striking price of the contract.

The Black-Scholes formula systematically gives higher values than those obtained with formula (1). On average, for the values of K, T, F and σ used in Table A2, the logarithmic difference is equal to 16 per cent; it is lower for higher values of the *dont* contract (high T, F and σ and low K), and vice-versa.

5. ESTIMATES OF VOLATILITY

5.1 The natural estimator

One of the hypotheses underlying formula (1) for the valuation of *dont* premium contracts is that the instantaneous rates of change of stock prices (dF/F) have a constant variance. It follows that the *natural* estimator of volatility is the standard deviation of the logarithmic differences of the stock prices:[29]

$$\hat{\sigma} = \sqrt{\sum_{j=1}^{m} \left[\ln\left(F_j / F_{j-1}\right) - \hat{\mu}\right]^2 / (m-1)}$$

where: F_j is the price observed at time j ($j = 1, ..., m$);

$$\hat{\mu} = \ln \frac{\left(\frac{F_m}{F_1}\right)}{m}.$$

Even if the continuous stochastic process that governs the changes in stock prices has constant parameters, the volatility found on the basis of prices observed at discrete intervals (days)[30] may di-

[26] For securities in backwardation (*riportati con deporto*) the (continuously compounded) *riporti* rate, which is due on *fine corrente* settlement, is equal, on an annual basis, to $r_{rip} = \ln[F_c / (F_c + dep)] / (\tau_2 - \tau_1)$, where F_c is the *prezzo di compenso* (which is usually equal to the official closing price observed on *riporti* day) and *dep* is the backwardation *(deporto)*. For securities in contango *(riportati con riporto)* the *riporti* rate, which is due on *fine prossimo* settlement, is equal to $r_{rip} = \ln(1 + i_{rip}/360)/(\tau_2 - \tau_1)$, where i_{rip} is the *riporti* rate fixed on the stock exchange on the basis of the convention of simple interest rates and the business year. For securities with a null contango (*riportati alla pari*), the *riporti* rate is equal to zero.

[27] Since the *riporti* rate r_{rip} is equal to the difference between r, the interest rate on money loans, and r_a, the interest rate on loans of securities, it follows that $\max(r_{rip}) = r$ since it is not possible to have $r_a < 0$.

[28] This percentage, which usually lies between 1 and 3 per cent for at-the-money *per fine corrente* premium contracts, is doubled in the case of *per fine prossimo* contracts.

[29] For an introduction to the problems of estimating historical volatility, see Cox and Rubinstein (1985; pp. 255-260 and 276-277).

[30] If daily data are used, the standard deviation needs to be annualized by multiplying it by $\sqrt{253}$ (since on average there are 253 trading days in a year) or, better, by $\sqrt{k365}$ where k is the ratio of trading to total days in the sample period.

verge significantly from the *true* volatility (Figlewski, 1989).[31]

In theory, increasing the frequency of the observations should improve the estimates.[32] Other problems may nonetheless arise: for example, the use of the intraday prices on the basis of which trades were actually carried out (transaction prices) would be likely to result in an overestimate of volatility, owing to the fluctuations in these prices within the bid-ask spread.[33] Furthermore, Cho e Frees (1988) have shown that the rounding of prices (typically to 1/8 of a dollar in the US market) result in estimates of volatility not necessarily improving as the frequency of observations during the day is increased.[34]

As regards the effect on prices of the detachment of dividends and rights, the series can be adjusted using suitable correction factors [Cox and Rubinstein (1985; p. 260)] or simply by discarding the ratios F_j/F_{j-1} in which F_j is an ex dividend or ex rights price. The second method is preferable for large samples, for which the loss of some observations is unimportant. The first method is, in fact, often vitiated by the fact that the correction factors use theoretical rather than market values. For example, in the case of dividends, the full price is determined by increasing the ex dividend price by the gross amount of the dividend, even though this may differ significantly from the market valuation of the dividend, above all for tax-related reasons.

In the light of the foregoing considerations, the analyses that follow are based on the application of the *natural* estimator to series of daily data missing of the observations corresponding to non-homogeneous prices. In fact, the standard deviation is estimated *within* each stock exchange month, thereby eliminating the effects of factors (deferment of settlement, detachment of dividends and rights) that are linked to the start of each stock exchange month.

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[31] In particular, Figlewski (1989) points out that «knowing that the true volatility is .15 only tells us that there is about 2/3 probability that a given month's volatility will be somewhere between .128 and .172». He also notes that «the sample volatility can differ from the true volatility only in discrete time, such as in the daily price series we are considering. If prices generated by a diffusion process could be followed continuously, the realized volatility over any finite time interval must equal the true volatility with probability 1.0».

[32] The *natural* estimator, $\hat{\sigma}^2$, follows a gamma distribution with a mean of σ^2 and variance $2\sigma^4/(m - 1)$. As the number of observations is increased, the variance of the estimator tends to zero, so that the sample estimate tends to its "true" value.

[33] Cox and Rubinstein therefore suggest using bid rather than actual prices.

[34] To eliminate, at least asymptotically, the distortions caused by rounding, Cho and Frees suggest using the following *temporal* volatility estimator in place of the *natural* estimator:

$$\hat{\sigma}^2 = \frac{2\hat{\mu}d}{\ln\left(d + \hat{\mu}\bar{\tau}_n\right) - \ln\left(d - \hat{\mu}\bar{\tau}_n\right)} \tag{1n}$$

where:

- d is the rounding constant, \$1/8 for example (Cho and Frees normalize the price series so as to have $S_0 = 1$ and put $d = d^*$, where $d^* = d/S_0$);
- τ_n = {first time $t > \tau_{n-1}$ such that $S(t)/S(\tau_{n-1}) \notin [(1 + d)^{-1}, (1 + d)]$}. In other words, τ_n is the first time at which a stock price increases or decreases by the multiple of $(1 + d)$ since the last price change. In particular, $\tau_0 = 0$.
- $\bar{\tau}_n = \tau_n/n$ (sample average of first-passage times);
- $\hat{\mu} = \ln[S(\tau_n)]/\tau_n$ (sample estimate of the process drift).

The simulation carried out by Cho and Frees shows that the *temporal* estimator is better for stocks with low and intermediate initial prices (respectively \$1 and \$25), but that the *natural* estimator is better for stocks with high initial prices (\$100).

Ball (1988) has also studied the effects of the rounding mechanism on the volatility of share prices. He notes, for example, that «for a security priced at ten dollars per share whose price dynamics are governed by Geometric Brownian Motion with an annual standard deviation of σ = .1587 and subject to rounding to the nearest eighth of a dollar, the resultant bias in variance estimation is 26.04 per cent» [Ball (1988; pp. 842)]. The estimator he recommends is:

$$\hat{\sigma}^2 = \frac{\sum_{j=1}^{m}\left[\ln\left(S_j / S_{j-1}\right)\right]^2}{m} - \frac{d^2}{6}. \tag{2n}$$

However, if intraday data are employed or the level of rounding is severe, the precision of estimates may be poor.

5.2 Autocorrelation

In reality, the assumption of constant volatility is no more than a useful approximation. A high degree of autocorrelation was noted by Fama as early as 1965: periods of pronounced stock price variability tend to alternate with periods of relative stability.

Accordingly, a mean-reversion has therefore been assumed, so that the current volatility is pushed towards its long-run average value, $\bar{\sigma}$, by an adjustment factor, γ, lying between 0 and 1:

$$\sigma_t - \sigma_{t-1} = \gamma(\bar{\sigma} - \sigma_{t-1}) + \varepsilon_t \tag{9}$$

The volatility change expected in the subsequent period ($\sigma_t - \sigma_{t-1}$) is proportional to the distance ($\bar{\sigma} - \sigma_{t-1}$) between the current volatility and its historical value. If γ is positive and $\bar{\sigma} > \sigma_{t-1}$ (current volatility less than its historical value), the expected change is positive, and vice-versa.

Putting $\beta_0 = \gamma\,\bar{\sigma}$ and $\beta_1 = 1 - \gamma$, equation (9) is rewritten as

$$\sigma_t = \beta_o + \beta_1 \quad \sigma_{t-1} + \varepsilon_t \tag{10}$$

and estimated using the daily series of a share price index (MIB *storico*) from January 20, 1975 to August 16, 1990 (186 stock exchange months).

The results of the regression based on ordinary least squares[35] are given in Table A3 equation (1). They show that there is significant autocorrelation. The current volatility tends towards its historical value $\bar{\sigma} = \beta_0/(1 - \beta_1) = .01$ (17.3 on an annual basis) as a result of the strong force of attraction exerted by the adjustment coefficient $\gamma = 1 - \beta_1 = .47$.

5.3 The relationship between volatility and rates of change

Another important empirical phenomenon observed in the US market is the negative correlation between the volatility and the rates of change in stock prices.[36] One possible theoretical explanation is that the reduction in prices results in an increase in firms' financial leverage (and probably in their operating leverage as well), thereby increasing price variability. This view underlies some option pricing models in which it is assumed that volatility is a function of stock prices (e.g. Cox and Ross [1976], Geske [1979] and Rubinstein [1983].

In these models the volatility of a stock is determined by the current price of the stock itself. However, empirical analyses of the US market conducted by Black [1976] showed that there is a lagged effect. This fact can be used in order to forecast future volatility.

Accordingly, the following equation is estimated:

$$\sigma_t = \beta_0 + \beta_1\,\sigma_{t-1} + \beta_2\,r_{t-1} + \varepsilon_t \tag{11}$$

where σ_t is the volatility observed in the stock exchange month t and r_{t-1} is the (continuously compounded) rate of change observed in the preceding month.

The results obtained do not suggest that the rate of change in the index observed in the preceding month goes far to explain volatility. The coefficient R^2 is only slightly higher than that obtained by estimating equation (10) and the sign of the coefficient β_2 is the opposite of the expected one (Table A3), equation 2.

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[35] An ARCH model has also been estimated, following Akgiray (1989). The identification, using the TR^2 test proposed by Engle (1982), made it appear advisable to estimate a second order ARCH, in view of a slight heteroschedasticity of the errors. The sum of the autoregressive coefficients estimated in the variance equation was not large (.36), thus confirming the weak heteroschedasticity of the errors. The results are in line with the tests made on the residuals of the regression based on ordinary least squares (DW = 2.14 and kurtosis coefficients very close to the theoretical value of the normal distribution).

[36] Two extreme examples of this negative correlation between volatility and stock prices are the depression of the 1930s and the stock exchange crisis of October 1987. In both cases the collapse of stock prices was accompanied by a period of considerable price volatility.

5.4 Implied volatility

Another way to estimate volatility is to use (1) to extract market expectations from the prices of premium contracts. The implied volatility, $\tilde{\sigma}$, is that value of σ for which the theoretical premium coincides with its market value. Given P_D, F_t, T, K and putting $P_D = \hat{P}_D$, formula (1) implies just one volatility, $\tilde{\sigma}$. The implied volatility is thus the volatility estimated by the market on the assumption that the actual premium is determined on the basis of (1).

Since the valuation formula proposed cannot be solved explicitly in terms of σ, it is necessary to employ numerical methods to calculate it. In this study we have used the well-known Newtonian algorithm defined by:

$$\begin{aligned}\sigma_{k+1} &= \sigma_k - \left(\hat{P}_D - P_D\right) / \frac{\partial \hat{P}_D}{\partial \sigma} = \\ &= \sigma_k \left[F_t N(d_1) - KN(d_2) - P_D\right] \;\; / \;\; \left[F_t N'(d_1)\sqrt{T}\right] \quad (K = 0,1,2....)\end{aligned} \tag{12}$$

where $\hat{P}_D$ and $\partial\hat{P}_D/\partial\sigma$ are given, respectively by (1) and (5). The iterative procedure starts with an arbitrary positive value, σ_0, and ends when $|\ln(\sigma_{k+1}/\sigma_k)| \leq \varepsilon$.

This method makes it possible to calculate the implied volatility of premium contracts with different maturities and derive the volatility expected by the market as a function of the time horizon. However, in practice premium contracts written on the same stock and for the same maturity tend to have rather different implied volatilities. This does not necessarily imply that (1) does not correctly represent the relationship between market premiums and determinant variables, but, at least in part, it reflects both the rounding implicit in the prices of premium contracts (which vary discretely) and the measurement errors of the variables. Specifically, as noted in Section 3, the information available on premium contracts suffers from asynchronous bias with respect to the corresponding closing prices of forward contracts.

One way to attenuate this bias is to combine, for each stock, the implied volatilities of contracts with different striking prices into a single estimate of future volatility. The simplest procedure is to use a weighted average of the single implied volatilities, and various weighting schemes have been proposed (Latanè and Rendleman [1976], Schmalensee and Trippi [l978], Patell and Wolfson [1979]). One frequently adopted scheme involves using the price elasticities of options with respect to volatility (Chiras and Manaster [1978]). This results in at-the-money contracts, with prices that are more sensitive to changes in volatility, being attributed a larger weight. More recent approaches to the estimation of volatility include the maximum-likelihood method proposed by Whaley [1982] and the unbiased-estimate procedure put forward by Butler and Schachter [1986].

Starting from the observation that the market prices of deep-out-of-the-money and deep-in-the-money options tend to differ systematically from the theoretical values, Macbeth and Merville [1979] and Beckers [1981] suggested that the implied volatility of at-the-money options (whose elasticity with respect to r is greater) provide forecasts of future volatility that are just as good as those obtained with more complex weighted averages.

Independently of the weighting scheme adopted, the empirical evidence regarding the US market shows that the forecasting power of implied volatilities is clearly superior with respect to historical volatilities. Latanè and Rendleman [1976] and Chiras and Manaster [1978] analyze the correlation between historical and implied volatilities, on the one hand, and ex post actual volatility, on the other, and conclude that implied volatility is clearly superior.

The forecasting power of implied volatilities in the Italian market is tested as follows. As a first step, the volatilities implied in the single *dont* contracts written on the last day of the 24 stock exchange months from September 1988 to August 1990 are calculated. A single implied volatility for each stock is then obtained as the weighted average of the single volatilities, using the elasticities of the premia with respect to r as weights. For each stock exchange month an "implied market volatility" is then calculated as the weighted average of the implied volatilities of the individual stocks, using the values of the premia traded as weights. Lastly, the following equation is estimated:

$$\sigma_t = \beta_0 + \beta_1 \sigma_{t-1} + \beta_2 r_{t-1} + \beta_3 \tilde{\sigma}_{t-1} + \varepsilon_t \tag{13}$$

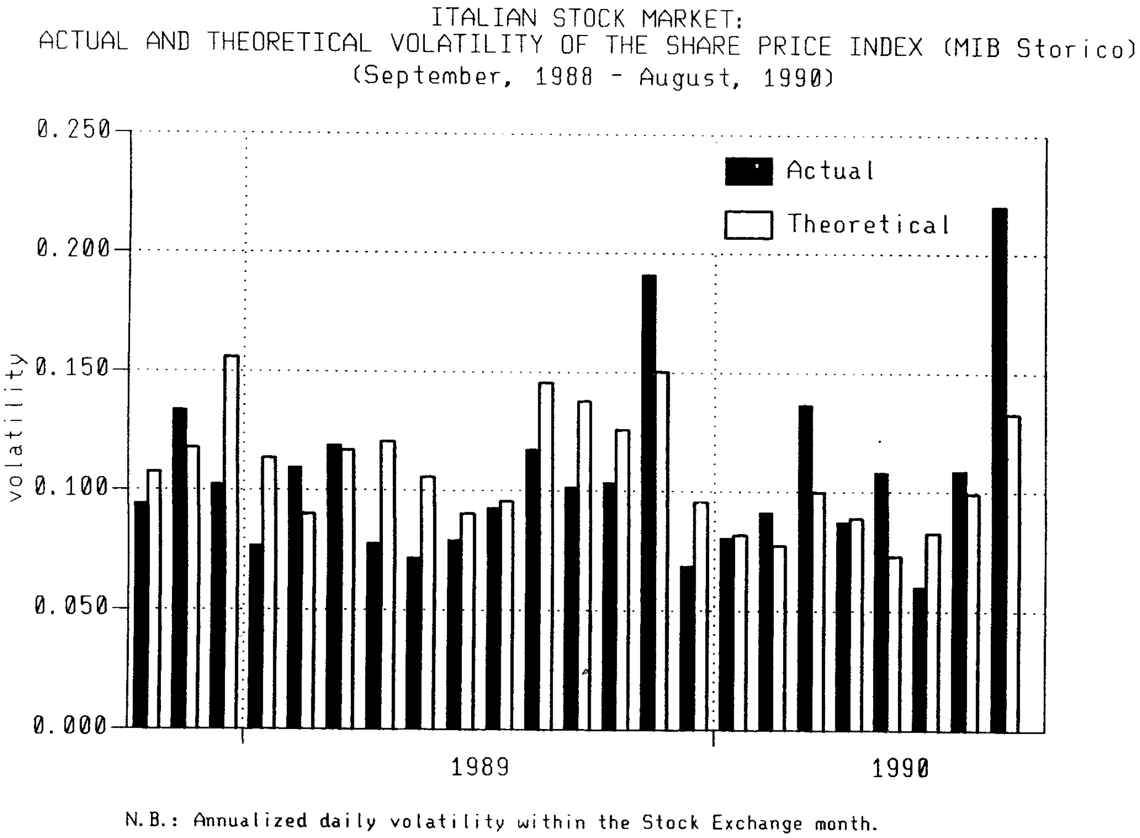

Figure 4

where σ_t is the volatility observed in the stock exchange month t, r_{t-1} is the (continuously compounded) rate of change observed in the previous and $\tilde{\sigma}_{t-1}$ is the implied volatility at the end of month $t-1$.

The results confirm those for the US market Table A3, equation (3). The implied volatility helps to explain the volatility actually observed in the subsequent period: an increase of one percentage point in the implied volatility at time $t-1$ becomes an increase of .44 points in the actual volatility at time t. The determination coefficient is found to be .41.

Figure 4 shows, for each stock exchange month, the variability observed in the share price index (MIB *storico*) and the forecast based on (13). The forecasts are sufficiently accurate, with a zero average absolute error and an average absolute percentage error of 24 per cent. The coefficient of correlation between the two series is equal to .64.

In order to obtain forecasts of the volatility of the individual stocks in the sample described in §3.1, equation (13) is estimated again with consideration being given indiscriminately to the stocks for which, in the first $t = 2 + j$ months ($j = 0, ..., 22$), the necessary information (σ_t, σ_{t-1}, r_{t-1}, $\tilde{\sigma}_{t-1}$) is available. Each of the 23 estimates is then made on the basis of a cross-section of stocks observed for an increasing number of months.

The results of the last regression, run on 526 observations regarding 23 stock exchange months, are given in Table A3 equation (4). They confirm, at the disaggregate level, the information content of implied volatilities, the coefficient of which (.17) is significantly different from zero (Student's t is equal to 4.4).

The average implied volatilities of the stocks underlying premium contracts on the Milan Stock Exchange, calculated for the sample period, are given in Appendix B, together with the historical

TABLE 5 Beta coefficients, historical and implied volatilities

Beta		*Historical volatilities*		*Implied volatilities*	
		MINIMUM VALUES			
Fisac di risp.	0.05	Saffa	0.14	Europa Metalli - LMI	0.15
Merloni	0.07	Valeo	0.14	Generali	0.17
Industrie Secco	0.09	Generali	0.14	Comp. Latina risp.	0.18
E. l'Espresso	0.09	Italcementi	0.14	Fiat priv.	0.19
Eliolona	0.10	Banco di Napoli	0.14	Fiat	0.20
		MAXIMUM VALUES			
Ifi priv.	1.91	B. Naz. Agr. priv.	0.48	Gerolimich	0.59
Saes	1.91	Mondadori risp.	0.42	Acqua Pia A.M.	0.48
Danieli & C. risp.	1.86	Alitalia	0.41	Linificio risp.	0.48
Pirelli S.p.A.	1.75	Terme Acqui risp.	0.41	SIM	0.46
Falck di risp.	1.72	Acqua Pia A.M. risp.	0.41	Sondel	0.45

volatilities and the beta coefficients.[37] Table 5 shows the extreme values of the three variables in the period in question.

6. ARBITRAGE OPPORTUNITIES

6.1 Actual and theoretical premia

The procedure described at the end of the previous section makes it possible to generate, at the end of a given stock exchange month, forecasts of the volatility of the stocks in the following month using only information that is already available.

On the basis of the forecasts and formula (1), it is possible to verify both the consistency of the actual and theoretical premia and the existence of arbitrage opportunities that operators failed to exploit.

The logarithmic difference between the actual and theoretical premia for the 11,501 *dont* contracts for which a forecast of the volatility is available averaged 16.3 per cent, with a standard deviation of 36.8. In two out of three cases the actual premia were larger than the theoretical values.

This difference may be due to operators not adapting the Black-Scholes formula to the features of premium contracts or to their using a higher volatility than the optimal forecast to take account of transaction costs.

The second of these hypotheses is tested by means of an extensive simulation designed to verify the existence, with transaction costs, of arbitrage opportunities that operators failed to exploit.

6.2 A simulation

Table A4 gives an example of how the replicating portfolio of a *dont* contract is created and subsequently rebalanced. The example refers to a *dont* contract on FIAT ordinary shares, with a striking price of 10,500 lire and maturity June 11, 1990. The average premium quotation was 135 lire on May 17, 1990, when the forward price was 10,285 lire.

To start with, the volatility of the FIAT shares was estimated for the June stock exchange month (15.6 per cent) and the theoretical value of the *dont* premium (83) lire then calculated using formula (1). Since the *dont* contract appeared to be overvalued, an arbitrage portfolio was created consisting of the sale of the *dont* contract itself and the purchase of the replicating portfolio.

[37] Consideration was given to the 290 stocks for which it was possible to calculate the historical volatility for each of the 24 stock exchange months in question.

The sale of a *dont* contract written on 10,000 FIAT shares at an average unit price of 135 lire results in a credit at settlement of 1.35 million lire. The theoretical value of the contract[38] is .52 million lire (1.35-.83), which falls to .387 million when account is taken of transaction costs (commissions[39] and stamp duties).

After calculating the hedge ratio, which indicates the number of stocks to buy, the replicating portfolio is created by buying 3,000 FIAT ordinary shares forward (3,000 ≈ 10,000 × .3134)[40] at a unit price of 10,285 lire, for a total consideration of 30.855 million lire. With transaction costs amounting to .108 (commissions, .077 and stamp duty, .031), the value of the forward contract[41] at the settlement date is negative and equal to .108 (0 – .108). The algebraic sum of the two contracts gives a value of the arbitrage contract equal to .279 (.387 – .108).

In order to maintain the equivalence between the *dont* contracts and the forward contract portfolio, it is necessary to adjust the stock of shares by buying or selling forward. For example, on May 18, 1990 the price of the share increased to 10,390 lire and the hedge ratio accordingly rose (to .4039), making it necessary to buy another 1,000 shares. When account is taken of the transaction costs incurred in the first two days (.144), the value of the forward contracts becomes .171 (41.560 – 41.245 – .144). The increase on the previous day is offset by the fall in the value of the *dont* contract sold.

Theoretically, the offset would be perfect under the assumptions on which formula (1) is based, notably the absence of transaction costs. In this case the initial value of the arbitrage portfolio would remain unchanged and be realized at the settlement date. In the foregoing example the existence of transaction costs, amounting to .884 (.631 + .253) at the expiration of the *dont* contract, means that the arbitrage operation closes not with a profit, but with a loss (–.192).

The result of the arbitrage operation described above would have been positive if the transaction costs had been curbed. One way of doing this in practice is to combine several arbitrage operations on *dont* contracts written on the same stock so as to eliminate the transactions of opposite sign.[42]

The foregoing led to an extended simulation being conducted involving all the *dont* contracts written between September 19, 1988 and August 16, 1990 on the ten stocks for which the most premium contracts were written (FIAT ordinary, Generali, Montedison, FIAT preferred, Olivetti, CIR, Mediobanca, Gemina, COMIT and IFI).[43] Of the 7,741 *dont* contracts analyzed, 856 were found to be out of line by more than 50 per cent compared with the theoretical values.[44] The simulation was therefore conducted on 856 *dont* contracts.

The *dont* contracts were bought or sold for quantities of stock such that the value of the underlying shares never exceeded 100 million lire (about 90,000 dollars); at the same time the replicating portfolio was sold or bought. The monthly average number of *dont* contracts in the arbitrage portfolio amounted to 58. On only 3 of the 505 days considered was the value of the arbitrage portfolio negative, with the largest negative value amounting to about 7 million lire. The profits earned using this strategy totaled 659 million lire. The monthly profits averaged 27 million, with a minimum of -7 million and a maximum of 120 million.

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[38] The value of the contract is the difference between the amount that will be credited at settlement (1.35 million lire) and the product of the theoretical unit value of the *dont* premium (83 lire) and the number of traded shares (10,000).

[39] It is assumed that the commission for forward sales is equal to 2.5 per thousand and that those for premium contracts (levied against the value of the underlying securities at the striking price) are half this amount (1.25 per thousand).

[40] The amount was rounded to take account of the size of the minimum lot (1,000 shares).

[41] The value of the forward contracts is equal to the difference between the current value of all the shares purchased and the cumulate of the values agreed in the individual trades.

[42] Another possibility is to rebalance the replicating portfolio whenever the theoretical hedge ratio differs from the current one by more than a fixed percentage, rather than every day.

[43] The only contracts excluded are those for which the original conditions changed in connection with operations affecting the companies' capital and/or the distribution of dividends.

[44] The mean of the logarithmic differences between the actual and the theoretical premia of this subsample was smaller than that of the whole sample (10.8 per cent as against 16.3 per cent).

In order to verify the absence of risk in the arbitrage portfolio, the profits are regressed on the monthly changes in the share price index.[45] The results of this test confirm the absence of linear correlation between the two variables: $R^2 = .01$, the intercept is equal to 28.3 million lire (Student's t is equal to 3.7), and the angular coefficient is not significantly different from zero (Student's t is equal to .4).

7. CONCLUSIONS

Premium contracts are special forms of options on forward contracts. Making assumptions similar to those of the Black-Scholes model, Barone and Cuoco (1988) showed that there exists a dynamic strategy (involving forward sales and purchases of stocks and bonds) able to replicate the final value of a premium contract. This makes it possible to determine the equilibrium premium, i.e. the premium that eliminates opportunities of arbitraging between a given premium contract and the replicating portfolio.

On the basis of a sample comprising all the *dont* contracts written on the Milan Stock Exchange in the 24 months from August 17, 1988 to August 16, 1990, an empirical test is made of the arbitrage opportunities offered by the market. To this end, the article describes certain regulatory aspects regarding the transaction costs involved in premium contracts and examines the question of forecasting volatility, the only parameter of the valuation formula that is not directly observable.

The results obtained show that forecasts of volatility based on historical volatility can be considerably improved by taking account of the current level of prices and, above all, of the market expectations implied in the current quotations of premium contracts (implied volatilities).

The analysis of the sample shows that there are percentage differences, which are sometimes extremely large, between the market premia and the corresponding equilibrium values and that these differences tend to be positive. One possible explanation put forward in this paper is that operators use the standard Black-Scholes formula, without adjusting it to take account of the special institutional features of the Italian market.

In the absence of transaction costs, these differences offer arbitrage opportunities. The simulation conducted on the sample confirms, moreover, that substantial arbitrage opportunities remain even after transaction costs are taken into account.

[45] There remains the risk of the quotation of the stock bought or sold under premium contracts being suspended, making it impossible to adjust the weights of the replicating portfolio.

APPENDIX A

TABLE A1 Forward and premium contracts on the Milan Stock Exchange (millions of lire)

Period	*Premium turnover*	*Securities turnover*		*Options market share (a)/(b)×100*
		options (a)	*forward (b)*	
1985	263,108	5,207,558	26,556,182	19.61
1986	707,209	10,602,254	66,660,844	15.91
1987	336,986	11,028,778	41,966,999	26.28
1988	362,954	11,384,736	41,269,230	27.59
1989	340,280	12,315,282	53,403,268	23.06
1990[2]	173,644	7,286,256	43,121,200	16.90
1988 - Jan.	14,857	494,447	1,699,093	29.10
Feb.	20,370	708,982	3,799,230	18.66
Mar.	42,966	1,135,411	5,606,323	20.25
Apr.	52,717	1,318,260	3,173,693	41.54
May	30,305	931,966	2,425,968	38.42
Jun.	21,480	784,386	3,341,154	23.48
Jul.	37,893	1,199,282	3,186,875	37.63
Aug.	22,369	768,837	2,868,777	26.80
Sep.	20,491	817,775	2,832,819	28.87
Oct.	35,273	1,190,829	5,568,694	21.38
Nov.	41,297	1,198,863	3,822,418	31.36
Dec.	22,936	835,698	2,944,186	28.38
1989 - Jan.	19,829	746,026	4,316,608	17.28
Feb.	24,004	976,534	2,907,206	33.59
Mar.	14,918	590,535	3,460,793	17.06
Apr.	21,721	781,182	3,480,498	22.44
May	16,297	713,239	3,190,775	22.35
Jun.	26,839	979,300	5,974,696	16.39
Jul.	53,585	1,581,497	6,138,663	25.76
Aug.	52,422	1,552,959	6,770,415	22.94
Sep.	42,398	1,370,441	5,441,000	25.19
Oct.	29,534	1,207,007	5,003,229	24.12
Nov.	19,833	839,346	3,314,113	25.33
Dec.	18,900	977,216	3,405,272	28.70
1990 - Jan.	12,599	535,848	5,306,900	10.10
Feb.	19,939	867,191	3,594,300	24.13
Mar.	15,331	688,112	3,978,300	17.30
Apr.	18,380	863,942	4,263,900	20.26
May	18,354	865,541	6,546,700	13.22
Jun.	28,276	1,179,956	6,392,900	18.46
Jul.	26,860	981,669	5,166,800	19.00
Aug.	14,776	607,784	5,210,100	11.67
Sep.	13,481	487,407	2,661,300	18.31

1. Calendar months and years.
2. For the first nine months only.
Source: The Stockbrokers' Executive Committee of the Milan Stock Exchange, *Il comportamento in Borsa dei valori azionari*, Milan 1989. The figures for 1990 are estimates.

TABLE A2 Theoretical values of dont premiums: comparison with the Black-Scholes formula

Volatility (σ)	Striking price (K)	*Fine corrente forward price* [$F(0, \tau_1)$]														
		900			950			1,000			1,050			1,100		
		BC	*BS*	*R*	*BC*	*BS*	*R*	*BC*	*BS*	*R*	*BC*	*BS*	*R*	*BC*	*BS*	*R*
Fine corrente premia (T = 30/365; τ_1 = 45/365)																
	900	21	24	2.3	55	61	0.5	101	108	0.1	150	157	0.0	200	207	0.0
	950	5	6	0.5	22	26	2.3	56	62	0.6	101	108	0.1	150	158	0.0
0.20	1,000	1	1	0.1	6	7	0.6	23	27	2.3	56	63	0.6	101	109	0.1
	1,050	0	0	0.0	1	1	0.1	6	8	0.6	24	28	2.3	57	64	0.7
	1,100	0	0	0.0	0	0	0.0	1	2	0.1	7	9	0.7	25	30	2.3
	900	41	45	4.6	72	77	2.4	111	117	1.2	155	161	0.5	202	209	0.2
	950	22	24	2.3	43	47	4.6	74	79	2.5	112	118	1.3	156	163	0.6
0.40	1,000	11	12	1.1	24	27	2.4	46	50	4.6	76	81	2.6	114	120	1.4
	1,050	5	5	0.4	12	14	1.2	26	29	2.5	48	52	4.6	78	84	2.7
	1,100	2	2	0.2	6	6	0.5	14	15	1.2	28	31	2.6	50	55	4.6
Fine prossimo premia (T = 60/365; τ_2 = 75/365; r_{rip} = 10%)																
	900	33	41	3.7	67	79	1.9	111	125	1.2	159	174	1.0	209	224	1.0
	950	13	18	1.4	35	44	3.7	69	81	2.0	112	126	1.3	160	175	1.1
0.20	1,000	4	6	0.4	15	20	1.5	37	46	3.7	71	83	2.1	113	128	1.3
	1,050	1	2	0.1	5	8	0.5	16	22	1.6	39	48	3.7	72	85	2.1
	1,100	0	0	0.0	1	2	0.1	6	9	0.6	18	24	1.6	40	51	3.7
	900	62	70	6.9	93	103	4.8	130	141	3.4	172	184	2.4	217	230	1.8
	950	41	47	4.3	66	74	6.9	97	106	4.9	133	145	3.5	174	187	2.5
0.40	1,000	26	30	2.6	44	51	4.4	69	77	6.9	100	110	5.0	136	148	3.6
	1,050	16	19	1.5	29	34	2.8	48	54	4.5	73	81	6.9	103	114	5.1
	1,100	9	11	0.8	18	22	1.7	32	37	2.9	51	58	4.6	76	85	6.9

TABLE A3 Explanatory variables of volatility

Equation	*Coefficients*				*DW*	R^2
	β_0	β_1	β_2	β_3		
1 $\sigma_t = \beta_0 + \beta_1 \sigma_{t-1} + \varepsilon_t$	0.005	0.534			2.13	0.29
	(6.4)	(8.5)				
2 $\sigma_t = \beta_0 + \beta_1 \sigma_{t-1} + \beta_2 r_{t-1} + \varepsilon_t$	0.004	0.569	0.014		2.13	0.31
	(6.0)	(9.1)	(2.7)			
3 $\sigma_t = \beta_0 + \beta_1 \sigma_{t-1} + \beta_2 r_{t-1} + \beta_3 \tilde{\sigma}_{t-1} + \varepsilon_t$	-0.001	0.070	-0.024	0.441	1.83	0.41
	(-0.3)	(0.3)	(-2.0)	(2.8)		
4 $\sigma_t = \beta_0 + \beta_1 \sigma_{t-1} + \beta_2 r_{t-1} + \beta_3 \tilde{\sigma}_{t-1} + \varepsilon_t$	0.112	0.225	-0.062	0.175	2.05	0.15
	(10.6)	(4.8)	(-1.4)	(4.4)		

1. Volatility (σ_t) of the MIB index in 186 stock exchange months (75/1/20 - 90/8/16).
2. Volatility (σ_t) and rate of change (r_t) of the MIB index in 186 stock exchange months (75/1/20 - 90/8/16).
3. Volatility (σ_t) and rate of change (r_t) of the MIB index in 24 stock exchange months (88/8/17 - 90/8/16) and average implied volatility ($\tilde{\sigma}_t$).
4. Volatility (σ_t), rate of change (r_t) and implied volatility ($\tilde{\sigma}_t$) of selected stocks in 24 stock exchange months (88/8/17 - 90/8/16; 526 observations).

TABLE A4 Arbitrage simulation: sale of a dont contract and purchase of the replicating portfolio

				PREMIUM PORTFOLIO				FORWARD PORTFOLIO											ARBITRAGE
Date	Dont Act.[2]	Dont Theor.[3]	Hedge Ratio[3]	Sale of dont		Transaction costs[6]	Net value (mil.)	Operations in shares[7]				Total shares		Commissions		Stamp Duty		Net value (mil.)	Balance (mil.)
				Cum.[4]	Value[5] (mil.)			Price	no.	Value (mil.)	Cumul. (mil.)	no.	Value (mil.)	-	Cumul.	-	Cumul.		
90/5/17	100-170	83	0.3134	1.35	0.520	132,650	0.387	10,285	3,000	30.855	30.855	3,000	30.855	77,138	77,138	30,900	30,900	-0.108	0.279
90/5/18	135-170	117	0.4039	1.35	0.176	132,650	0.043	10,390	1,000	10.39	41.245	4,000	41.56	25,975	103,112	10,400	41,300	0.171	0.214
90/5/21	130-210	110	0.4043	1.35	0.247	132,650	0.114	10,398	-	-	41.245	4,000	41.592	-	103,112	-	41,300	0.203	0.317
90/5/22	140-190	191	0.5793	1.35	-0.560	132,650	-0.693	10,570	2,000	21.14	62.385	6,000	63.42	52,850	155,962	21,200	62,500	0.817	0.124
90/5/23	110-160	149	0.5071	1.35	-0.141	132,650	-0.274	10,500	-1,000	-10.5	51.885	5,000	52.5	26,250	182,212	10,500	73,000	0.36	0.086
90/5/24	130-165	145	0.5069	1.35	-0.101	132,650	-0.234	10,500	-	-	51.885	5,000	52.5	-	182,212	-	73,000	0.36	0.126
90/5/25	150-170	162	0.5516	1.35	-0.272	132,650	-0.405	10,540	1,000	10.54	62.425	6,000	63.24	26,350	208,562	10,600	83,600	0.523	0.118
90/5/28	100-130	106	0.4501	1.35	0.285	132,650	0.152	10,455	-1,000	-10.455	51.97	5,000	52.275	26,138	234,700	10,500	94,100	-0.024	0.129
90/5/29	100-130	117	0.4990	1.35	0.181	132,650	0.048	10,487	-	-	51.97	5,000	52.435	-	234,700	-	94,100	0.136	0.185
90/5/30	140-170	182	0.6489	1.35	-0.471	132,650	-0.604	10,610	1,000	10.61	62.58	6,000	63.66	26,525	261,225	10,700	104,800	0.714	0.110
90/5/31	-	230	0.7449	1.35	-0.949	132,650	-1.082	10,685	1,000	10.685	73.265	7,000	74.795	26,713	287,937	10,700	115,500	1.127	0.045
90/6/1	230	277	0.8229	1.35	-1.420	132,650	-1.553	10,750	1,000	10.75	84.015	8,000	86	26,875	314,812	10,800	126,300	1.544	-0.009
90/6/4	-	300	0.8975	1.35	-1.655	132,650	-1.788	10,789	1,000	10.789	94.804	9,000	97.101	26,973	341,785	10,800	137,100	1.818	0.031
90/6/5	-	276	0.8955	1.35	-1.408	132,650	-1.541	10,765	-	-	94.804	9,000	96.885	-	341,785	-	137,100	1.602	0.061
90/6/6	-	130	0.6832	1.35	0.049	132,650	-0.084	10,590	-2,000	-21.18	73.624	7,000	74.13	52,950	394,735	21,200	158,300	-0.047	-0.130
90/6/7	-	185	0.8393	1.35	-0.499	132,650	-0.632	10,670	1,000	10.67	84.294	8,000	85.36	26,675	421,410	10,700	169,000	0.476	-0.155
90/6/8	-	124	0.7551	1.35	0.111	132,650	-0.022	10,602	-	-	84.294	8,000	84.816	-	421,410	-	169,000	-0.068	-0.090
90/6/11	-	0	0	1.35	1.350	132,650	1.217	10,471	-8,000	-83.768	0.526	0	0	209,420	630,830	83,800	252,800	-1.409	-0.192

1. The simulation refers to a dont contract written on 10,000 FIAT ordinary shares with a striking price of 10,500 lire and *fine corrente* maturity, in the period from May 17, 1990 to *risposta premi* day (June 11, 1990). The values given refer to the settlement date (June 28, 1990).
2. The minimum and maximum actual prices, observed on each trading day.
3. Calculated using formula [1].
4. Assuming the sale of a dont contract written on 10,000 FIAT ordinary shares at the average observed price of 135 lire (the dont effectively written referred to 1,550,000 shares). The amount indicated is the amount that will be credited at settlement.
5. The theoretical value of the *dont* contract sold [(135-Dont theoretical)x10,000] is positive initially since the implied volatility (21.65) is greater than that forecast (15.60). The effective volatility is equal to 13.62.
6. Commissions (131,250=0.00125x10,500x10,000) plus taxes (1,400=.001x135x10,000).
7. Purchases are shown with a plus sign, sales with a minus sign. The amounts corresponding to the hedge ratio have been rounded to take account of the size of the minimum lot (1,000 shares).

APPENDIX B

TABLE B1 Shares listed on the Milan Stock Exchange (August 17, 1988 - August 16, 1990)

Share	*Historical Beta*	*Historical Volatility*	*Implied Volatility*	*Sector*
Abb-Tecnomasio	0.48	0.22		Electrical
Acq. De Ferrari G.	0.83	0.27		Sundry
Acq. De Ferrari G. savings n.c.	0.71	0.36		Sundry
Acqua Pia Antica Marcia	0.36	0.30	0.48	Financial
Acqua Pia Antica Marcia savings n.c.	0.36	0.41		Financial
Acque Potabili	0.70	0.24		Sundry
Aedes	0.91	0.18		Property and building
Aedes savings n.c.	0.98	0.23		Property and building
Aeritalia	1.38	0.15	0.22	Engineering and motor vehicles
Alitalia	0.61	0.41		Communications
Alitalia preferred	1.08	0.25		Communications
Alivar	0.77	0.27		Food and agricultural products
Alleanza Assicurazioni	0.84	0.18	0.28	Insurance
Alleanza savings n.c.	0.65	0.18		Insurance
Ansaldo Trasporti	0.93	0.17		Electrical
Assitalia	0.96	0.21	0.27	Insurance
Attività immobiliari	0.98	0.19	0.37	Property and building
Auschem	0.96	0.29		Chemical, oil and rubber products
Auschem savings n.c.	0.71	0.28		Chemical, oil and rubber products
Ausiliare	0.30	0.19		Communications
Ausonia Assicurazioni	0.63	0.21	0.21	Insurance
Autostrada TO-MI	1.16	0.24		Communications
Autostrade C. and C. preferred	0.56	0.15		Communications
AVIR Finanziaria	0.79	0.15	0.31	Financial
B.ca Naz. dell'Agric. savings n.c.	0.24	0.22		Banking
B.ca Naz. dell'Agric. preferred	1.36	0.48	0.39	Banking
B.co di Chiavari e della Riv. Ligure	1.04	0.23		Banking
Banca Agricola Milanese	0.60	0.28		Banking
Banca Commerciale Italiana	0.77	0.20	0.23	Banking
Banca Commerciale Italiana savings n.c	. .54	0.17		Banking
Banca Manusardi & C.	0.75	0.26	0.38	Banking
Banca Mercantile Italiana	0.81	0.24		Banking
Banca Naz. dell'Agricoltura	0.55	0.34	0.32	Banking
Banca Toscana	0.75	0.20	0.26	Banking
Banco Ambrosiano Veneto	0.93	0.22	0.37	Banking
Banco Ambrosiano Veneto savings n.c.	0.70	0.32		Banking
Banco di Napoli (savings parts) n.c.	0.33	0.14		Banking
Banco di Roma	0.83	0.25	0.29	Banking
Banco di Sardegna savings n.c.	0.49	0.20		Banking
Banco Lariano	0.67	0.21		Banking
Bastogi	0.11	0.23	0.29	Financial
Benetton Group	0.94	0.22	0.28	Textiles
Boero Bartolomeo	0.98	0.28		Chemical, oil and rubber products
Bonifiche Ferraresi	0.55	0.19		Food and agricultural products
Bonifiche Siele	1.49	0.31		Financial
Bonifiche Siele savings n.c.	0.72	0.32		Financial

Share	*Historical Beta*	*Historical Volatility*	*Implied Volatility*	*Sector*
Brioschi	1.37	0.30		Financial
Burgo	1.31	0.17	0.24	Paper and publishing
Burgo savings n.c.	1.08	0.19		Paper and publishing
Burgo preferred	1.16	0.22		Paper and publishing
Buton	0.70	0.24		Financial
C.A.L.P. "Crist. Art. La Piana"	0.56	.	26.00	Chemical, oil and rubber products
Caffaro	1.51	0.20	0.36	Chemical, oil and rubber products
Caffaro savings	1.31	0.22		Chemical, oil and rubber products
Calcestruzzi	0.40	0.19	0.28	Property and building
Caltagirone	0.57	0.25		Property and building
Caltagirone savings n.c.	0.70	0.41		Property and building
Cam-Fin.	1.16	0.27		Financial
Cantieri Metallurgici It.	0.72	0.20		Financial
Cantoni I.T.C.	0.48	0.24		Textiles
Cartiera di Ascoli	0.72	0.31		Paper and publishing
Cartiere Sottrici Binda	0.64	0.22		Paper and publishing
Cementeria di Augusta	0.94	0.18		Cement, ceramics and building materials
Cementeria di Barletta	0.77	0.24		Cement, ceramics and building materials
Cementeria di Merone	0.82	0.22		Cement, ceramics and building materials
Cementeria di Merone savings n.c.	1.03	0.26		Cement, ceramics and building materials
Cementerie di Sardegna	1.02	0.19		Cement, ceramics and building materials
Cementerie Siciliane	0.87	0.16		Cement, ceramics and building materials
Cementir	1.04	0.18		Cement, ceramics and building materials
Ciga	1.27	0.23		Sundry
Ciga savings n.c.	1.68	0.23		Sundry
Cir	1.16	0.20	0.21	Financial
Cir savings n.c.	1.52	0.23	0.27	Financial
Cir risp.	1.06	0.21		Financial
Cofide	0.64	0.20	0.24	Financial
Cofide savings n.c.	1.22	0.24	0.31	Financial
Cogefar Impresit	0.84	0.18		Property and building
Cogefar Impresit risp. n.c.	1.23	0.25		Property and building
Comau Finanziaria	1.35	0.25		Financial
Comp. Ass. Milano	0.73	0.20		Insurance
Comp. Ass. Milano savings n.c.	0.56	0.25		Insurance
Comp. Latina Ass.	0.79	0.21	0.25	Insurance
Comp. Latina Ass. risp. n.c.	0.75	0.25	0.18	Insurance
Credito Commerciale	0.57	0.19		Banking
Credito Fondiario	0.71	0.24		Banking
Credito Italiano	1.15	0.23	0.29	Banking
Credito Italiano savings n.c.	0.91	0.22		Banking
Credito Lombardo	0.55	0.24		Banking
Credito Varesino	0.54	0.21	0.32	Banking
Credito Varesino savings n.c.	0.56	0.25		Banking
Cucirini	0.60	0.34	0.35	Textiles
Dalmine	0.69	0.29	0.43	Mineral products and basic metals
Danieli & C.	1.61	0.20		Engineering and motor vehicles
Danieli & C. savings n.c.	1.86	0.21		Engineering and motor vehicles
Dataconsyst	1.60	0.20	0.41	Engineering and motor vehicles

Share	*Historical Beta*	*Historical Volatility*	*Implied Volatility*	*Sector*
Del Favero	1.01	0.21		Property and building
E. l'Espresso	0.09	0.25		Paper and publishing
Editoriale	0.82	0.22		Financial
Eliolona	0.10	0.22		Textiles
Eridania	1.28	0.20	0.36	Food and agricultural products
Eridania savings n.c.	1.13	0.19		Food and agricultural products
Euromobiliare	0.39	0.18		Financial
Euromobiliare savings n.c.	0.64	0.30		Financial
Europa Metalli - LMI	1.29	0.22	0.15	Mineral products and basic metals
Fabbrica Milanese Conduttori	0.10	0.26		Chemical, oil and rubber products
Faema	0.89	0.29		Engineering and motor vehicles
Falck	1.15	0.23	0.28	Mineral products and basic metals
Falck savings	1.72	0.28		Mineral products and basic metals
Ferruzzi Agricola savings n.c.	1.41	0.21		Financial
Ferruzzi Agricola Fin.	1.25	0.20	0.29	Financial
Ferruzzi Agricola Fin. savings	0.63	0.25		Financial
Ferruzzi Fin.	0.79	0.17	0.22	Financial
Fiar	0.68	0.23		Engineering and motor vehicles
Fiat	1.24	0.15	0.20	Engineering and motor vehicles
Fiat savings n.c.	1.23	0.16	0.28	Engineering and motor vehicles
Fiat preferred	1.34	0.17	0.19	Engineering and motor vehicles
Fidenza Vetraria	1.27	0.20		Chemical, oil and rubber products
Fidis	1.34	0.19	0.30	Financial
Fimpar	0.61	0.18		Financial
Fimpar savings n.c.	1.39	0.25	0.35	Financial
Finanziaria Centro Nord	0.25	0.28		Financial
Finanziaria Pozzi-Ginori	0.34	0.24		Financial
Finanziaria Pozzi-Ginori savings n.c.	0.46	0.24		Financial
Finarte	0.11	0.19	0.30	Financial
Finrex	0.31	0.35		Financial
Finrex savings	0.74	0.33		Financial
Fisac	0.40	0.36		Textiles
Fisac savings	0.05	0.35		Textiles
Fiscambi Holding	0.41	0.37		Financial
Fiscambi Holding savings n.c.	0.52	0.33		Financial
Fochi Filippo	1.08	0.25		Engineering and motor vehicles
Fornara	0.58	0.21	0.29	Financial
Franco Tosi	0.80	0.22		Engineering and motor vehicles
Gemina	1.35	0.22	0.24	Financial
Gemina savings n.c.	1.39	0.27	0.41	Financial
Generali	0.76	0.14	0.17	Insurance
Gerolimich	0.49	0.27	0.59	Financial
Gerolimich savings n.c.	0.47	0.23		Financial
Gilardini	1.57	0.21	0.26	Engineering and motor vehicles
Gilardini savings n.c.	1.09	0.21		Engineering and motor vehicles
Gim	0.70	0.23		Financial
Gim savings n.c.	1.30	0.26		Financial
Grassetto	1.26	0.22		Property and building
Gruppo Editoriale Fabbri preferred	1.14	0.20		Paper and publishing

Share	*Historical Beta*	*Historical Volatility*	*Implied Volatility*	*Sector*
Ifi preferred	1.91	0.21	0.26	Financial
Ifil	1.34	0.21	0.36	Financial
Ifil savings n.c.	1.55	0.23		Financial
Immobiliare Metanopoli	0.68	0.20	0.29	Property and building
Industrie Secco	0.09	0.40		Engineering and motor vehicles
ISEFI	0.33	0.25	0.27	Financial
Italcable	1.65	0.23		Communications
Italcable savings n.c.	1.23	0.22		Communications
Italcementi	1.09	0.14		Cement, ceramics and building materials
Italcementi savings n.c	1.28	0.24	0.22	Cement, ceramics and building materials
Italgas	0.69	0.18	0.26	Chemical, oil and rubber products
Italia Assicurazioni	0.97	0.24	0.41	Insurance
Italmobiliare	1.06	0.16	0.25	Financial
Italmobiliare savings n.c.	1.68	0.26		Financial
Jolly Hotels	0.72	0.19	0.34	Sundry
Jolly Hotels savings n.c.	0.70	0.27		Sundry
Kernel Italiana	0.47	0.28		Financial
L'Abeille italiana	0.88	0.16		Insurance
La Fondiaria	0.84	0.17	0.23	Insurance
La Previdente	1.11	0.22		Insurance
La Rinascente	0.99	0.19	0.33	Wholesale and retail trades
La Rinascente savings n.c.	1.04	0.18		Wholesale and retail trades
La Rinascente preferred	1.10	0.20	0.39	Wholesale and retail trades
Linificio Canapif. Naz. savings n.c.	0.33	0.18	0.48	Textiles
Linificio Canapificio Naz.	0.90	0.22		Textiles
Lloyd Adriatico	1.13	0.21	0.38	Insurance
Lloyd Adriatico risp. n.c.	0.90	0.25		Insurance
Maffei	0.18	0.24		Mineral products and basic metals
Magneti Marelli	1.57	0.20		Engineering and motor vehicles
Magneti Marelli savings	1.46	0.21		Engineering and motor vehicles
Magona	0.24	0.33		Minerarie-Metallurgiche
Manuli Cavi	0.64	0.20		Chemical, oil and rubber products
Manuli Cavi savings n.c.	0.54	0.20		Chemical, oil and rubber products
Marangoni	0.88	0.19		Chemical, oil and rubber products
Marzotto	0.78	0.17		Textiles
Marzotto savings n.c.	0.59	0.27		Textiles
Marzotto risp. port.	0.54	0.30		Textiles
Mediobanca	1.25	0.19	0.25	Banking
Merloni Elettrodomestici	0.07	0.20		Engineering and motor vehicles
Mira Lanza	0.43	0.17		Chemical, oil and rubber products
Mittel	0.62	0.23		Financial
Mondadori risp. n.c.	0.16	0.42		Paper and publishing
Montedison	1.17	0.20	0.24	Chemical, oil and rubber products
Montedison savings n.c.	1.68	0.23	0.30	Chemical, oil and rubber products
Montefibre	0.85	0.25	0.33	Chemical, oil and rubber products
Montefibre savings n.c.	0.28	0.21		Chemical, oil and rubber products
Necchi	0.78	0.24		Engineering and motor vehicles
Necchi savings n.c.	0.27	0.34		Engineering and motor vehicles
Nuovo Pignone	0.62	0.15		Engineering and motor vehicles

Share	*Historical Beta*	*Historical Volatility*	*Implied Volatility*	*Sector*
Olcese Veneziano	0.86	0.29		Textiles
Olivetti ord.	0.87	0.15	0.23	Engineering and motor vehicles
Olivetti preferred	0.93	0.25		Engineering and motor vehicles
Olivetti risp. n.c.	0.82	0.21		Engineering and motor vehicles
Pacchetti	0.82	0.23	0.29	Sundry
Partecipazioni Fin. savings n.c. ex W.	1.03	0.22		Financial
Partecipazioni Finanziarie ind.	0.55	0.19		Financial
Perlier	0.79	0.29		Chemical, oil and rubber products
Pierrel	1.15	0.29		Chemical, oil and rubber products
Pierrel savings n.c.	0.49	0.33		Chemical, oil and rubber products
Pininfarina	0.57	0.16		Engineering and motor vehicles
Pininfarina savings	0.72	0.19		Engineering and motor vehicles
Pirelli & C.	1.51	0.19	0.25	Financial
Pirelli & C. di ris. n.c.	1.12	0.21		Financial
Pirelli S.p.A.	1.75	0.21	0.28	Chemical, oil and rubber products
Pirelli S.p.A. savings	1.49	0.25		Chemical, oil and rubber products
Pirelli S.p.A. savings n.c.	1.09	0.22		Chemical, oil and rubber products
Poligrafici Editoriale	0.48	0.14	0.30	Paper and publishing
Raggio di Sole	0.45	0.19		Financial
Raggio di Sole savings	0.47	0.17		Financial
Ras	1.06	0.20	0.25	Insurance
Ras savings n.c.	1.43	0.23	0.39	Insurance
Recordati	0.93	0.17		Chemical, oil and rubber products
Recordati Risp. n.c.	0.93	0.17		Chemical, oil and rubber products
Rejna	0.79	0.19		Engineering and motor vehicles
Risanamento	0.47	0.20		Property and building
Risanamento savings n.c.	0.69	0.22		Property and building
Riva Finanziaria	0.57	0.22		Financial
Rodriquez	0.88	0.23		Engineering and motor vehicles
Rotondi	0.57	0.35		Textiles
Saes	1.91	0.24		Financial
Saes savings n.c.	1.49	0.22		Financial
Saes Getters preferred	0.88	0.21	0.27	Electrical
Saffa	1.00	0.14	0.23	Chemical, oil and rubber products
Saffa risp. port.	0.69	0.16		Chemical, oil and rubber products
Saffa savings n.c.	0.69	0.15		Chemical, oil and rubber products
Safilo	1.18	0.22		Engineering and motor vehicles
Safilo risp. port.	1.33	0.40		Engineering and motor vehicles
Sai	0.88	0.17	0.27	Insurance
Sai savings n.c.	1.38	0.23		Insurance
Saiag	0.36	0.20		Chemical, oil and rubber products
Saiag Risp. n.c.	1.33	0.28		Chemical, oil and rubber products
Saipem	0.75	0.27		Engineering and motor vehicles
Saipem savings	0.92	0.40		Engineering and motor vehicles
Sasib	1.00	0.20		Engineering and motor vehicles
Sasib savings n.c.	1.47	0.19		Engineering and motor vehicles
Sasib preferred	0.96	0.32		Engineering and motor vehicles
Schiapparelli 1824	1.08	0.21		Financial
Selm	0.12	0.22	0.27	Electrical

Share	*Historical Beta*	*Historical Volatility*	*Implied Volatility*	*Sector*
Selm savings n.c.	0.17	0.35		Electrical
Serfi	0.80	0.19		Financial
Setemer	1.41	0.24		Financial
Sifa	0.64	0.25		Financial
Sifa savings n.c.	0.52	0.27		Financial
SIM - Soc. It. Manufatti	0.13	0.28	0.46	Textiles
Sip	1.16	0.20	0.27	Communications
Sip savings n.c.	0.74	0.17	0.24	Communications
Sirti	1.12	0.18		Communications
SISA - Soc. Imb. spec.	0.76	0.29		Financial
Sme	0.84	0.21	0.30	Financial
Snia BPD	1.62	0.24	0.26	Chemical, oil and rubber products
Snia BPD savings	1.49	0.27		Chemical, oil and rubber products
Snia BPD savings n.c.	1.24	0.21		Chemical, oil and rubber products
Snia Fibre	0.84	0.19		Chemical, oil and rubber products
Snia Tecnopolimeri	0.97	0.19		Chemical, oil and rubber products
SO.PA.F.	0.62	0.19		Financial
SO.PA.F. savings n.c.	0.83	0.21		Financial
Sogefi	0.77	0.18		Financial
Sondel - Soc. Nordelettrica	0.23	0.23	0.45	Electrical
Sorin Biomedica	0.83	0.17		Chemical, oil and rubber products
Standa	0.55	0.18		Wholesale and retail trades
Standa savings n.c.	0.63	0.32		Wholesale and retail trades
Stefanel	1.16	0.22		Textiles
Tecnost	0.82	0.18		Engineering and motor vehicles
Teknecomp	1.04	0.22		Engineering and motor vehicles
Teknecomp risp. n.c.	0.91	0.17		Engineering and motor vehicles
Terme Acqui	1.19	0.28		Financial
Terme Acqui savings n.c.	0.77	0.41		Financial
Toro Ass.	1.36	0.22	0.42	Insurance
Toro Ass. preferred	1.28	0.25	0.38	Insurance
Toro Ass. risp. n.c.	1.41	0.22		Insurance
Trenno	1.21	0.23	0.31	Financial
Tripcovich & C.	0.86	0.26		Financial
Tripcovich & C. savings n.c.	0.88	0.29		Financial
Unicem	1.23	0.20		Cement, ceramics and building materials
Unicem savings n.c.	1.25	0.17		Cement, ceramics and building materials
Unione Subalpina di Assicurazioni	1.20	0.27		Insurance
Unipol preferred	0.96	0.19	0.30	Insurance
Valeo	1.15	0.14		Engineering and motor vehicles
Vetrerie Italiane Vetri	0.62	0.19		Chemical, oil and rubber products
Vianini Ind.	0.82	0.28		Property and building
Vianini Lavori	0.99	0.22		Property and building
Vittoria Assicurazioni	0.86	0.19		Insurance
Wabco Westinghouse	0.22	0.29		Engineering and motor vehicles
Worthington	0.83	0.30		Engineering and motor vehicles
Zignago	0.77	0.19	0.24	Food and agricultural products
Zucchi	1.32	0.21		Textiles

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